

Homework 2

Complex Numbers

- Compute the magnitude, phase (in degrees, $-180^\circ \leq \phi \leq 180^\circ$), real part, and imaginary part of the following expressions for $z_1 = 1 + j$, $z_2 = -1 - 2j$. You can use Matlab functions `abs()`, `phase()` or `atan2d()`, `real()`, and `imag()`.
 - z_1
 - z_2
 - $z_1 + z_2$
 - $z_1 z_2$
 - z_1 / z_2
 - $z_1 \bar{z}_1$
- For parts c, d, e, and f in the last problem, draw vectors representing z_1 , z_2 , and the final expression in each part. Describe how these operations work in terms of translation or rotation. Hint: for multiplication or division, consider the exponential form $z = Me^{j\phi}$ for insight. You can use Matlab or plot by hand.
- Plot the following parametric expressions in the complex plane (x-axis – real part, y-axis – imaginary part) for $-\infty < w < \infty$. You can use Matlab.
 - $z = w^2 + \frac{2}{w}j$
 - $z = \frac{jw}{1+jw}$
- Prove the following two statements, assuming z_1 and z_2 are arbitrary complex numbers of the form $z_i = a_i + b_i j$

$$|z_1 z_2|^2 = |z_1|^2 |z_2|^2$$

$$\left| \frac{z_1}{z_2} \right|^2 = \frac{|z_1|^2}{|z_2|^2}$$

Briefly describe how this can be extended to show

$$\left| \frac{z_1 z_2 z_3 \dots}{z_{10} z_{11} z_{12} \dots} \right|^2 = \frac{|z_1|^2 |z_2|^2 |z_3|^2 \dots}{|z_{10}|^2 |z_{11}|^2 |z_{12}|^2 \dots}$$

- Using Matlab, plot this parametric function in the complex plane for $-\infty < w < \infty$:

$$z = \frac{(3 + wj)(4 + wj)}{(1 + wj)(10 + wj)(2 + wj)}$$

- For the same function in the previous problem,
 - Write an analytical expression for the magnitude $M(w)$ by hand using the results of Problem 4. Note – there is no need to simplify (expand) this expression.
 - Write an analytical expression for the phase $\phi(w)$ by hand. Note – there is no need to simplify this expression; leave it in the form

$$\tan^{-1} x_1 + \tan^{-1} x_2 \dots$$
 - Plot $M(w)$ vs. w and $\phi(w)$ vs w in Matlab (hint: try `semilogx` for a logarithmic x axis).
- Consider a set of points that form a closed polygon in the complex plane when connected in order. For example:

$$1 + 0j, 2 + 0j, 2 + 1j, 1 + 1j, 1 + 0j$$

You wish to rotate these points by 30° about the origin in the complex plane. What complex number should you multiply by? Show this works using Matlab.

Newton's Method

Newton's method can be used to find the roots of a function. One starts with an initial guess x_0 . The guess is updated as

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

8. Consider the polynomial

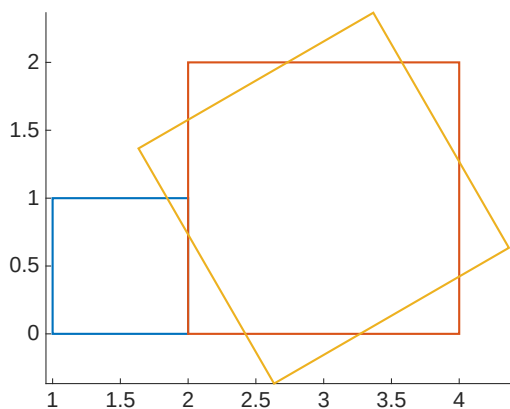
$$f(x) = x^3 + 1$$

- Use the Matlab command `roots()` to find the roots of the polynomial (p_1, p_2, p_3) .
- Analytically compute $f'(x)$.
- Pick a starting value $x_{0,1}$ from the complex plane and show it converges to root p_1 using Newton's method. Similarly show different starting values $x_{0,2}$ and $x_{0,3}$ converge to p_2 and p_3 , respectively. Plot each sequence of values $x_{n,i}$ in the complex plane and show they terminate at p_i .

Optional Bonus Problems:

- You wish to first scale the shape by factor α and then rotate it about an arbitrary point $c_1 + c_2j$ by θ degrees. Describe your methodology (equations) for doing this using complex algebra, and show it works in Matlab for $\alpha = 2$, $c_1 = 3$, $c_2 = 1$, $\theta = 30^\circ$.

Example output:



10. Consider the polynomial

$$f(x) = x^3 + 1$$

And a square in the complex plane $S = a + bj$, where $-1 \leq a \leq 1$ and $-1 \leq b \leq 1$

- Using Newton's method, map regions of S to the root they terminate at. Hint: create a mesh of points in S that represent starting points, and run Newton's

method on each of these points. You may want to vectorize the calculations. There may be some points that never converge, which is a known limitation to Newton's method. For these points, you can ignore them or return the pole as NaN.

- b. Make a plot in the complex plane showing these regions. For example, you could use 3 colors representing the 3 poles and shade S with these colors. Hint – Matlab function `contourf()` may be useful.
- c. Try your code on other polynomials of your choice!